ToM-Driver: Beyond Kinematic Models toward Social Intent Estimation with Theory-of-Mind in Autonomous Driving

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Introduction

- Autonomous driving requires understanding the intentions of surrounding vehicles in complex traffic environments. Inspired by the human cognitive concept of **Theory of Mind (ToM)**, this work proposes **ToM-Driver**, a framework that enables autonomous agents to reason about other drivers' intentions using large language models.
- **Keywords**: Theory of Mind (ToM); Autonomous Driving; Large Language Models (LLMs); Intent Inference

Conclusions

- We present **ToM-Driver**, a framework that integrates Theory of Mind reasoning into autonomous driving.
- By enabling agents to infer other vehicles' intentions, the system improves **decision-making and safety** in multi-agent scenarios, demonstrating the potential of cognitive reasoning approaches for future autonomous driving systems.

References

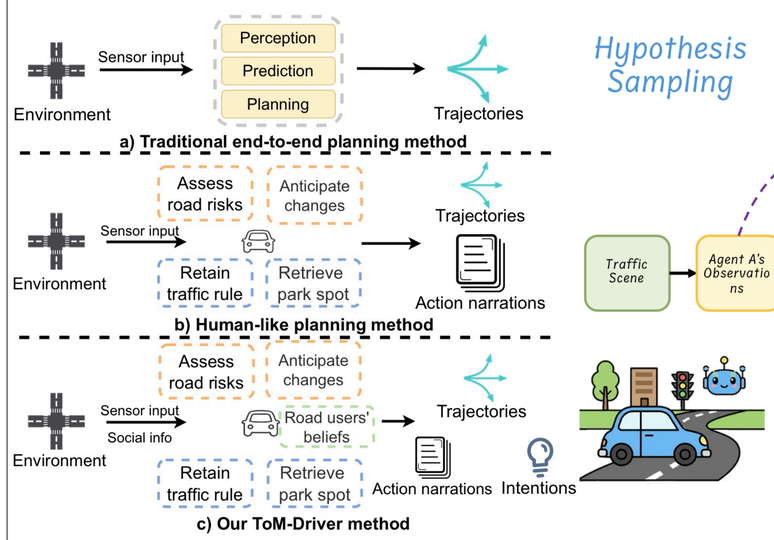
[1] Chris L. Baker, Rebecca Saxe, and Joshua B. Tenenbaum. Action understanding as inverse planning. Cognition, 113(3):329–349, 2009.

[2] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuscenes: A multi-modal dataset for autonomous driving, 2020.

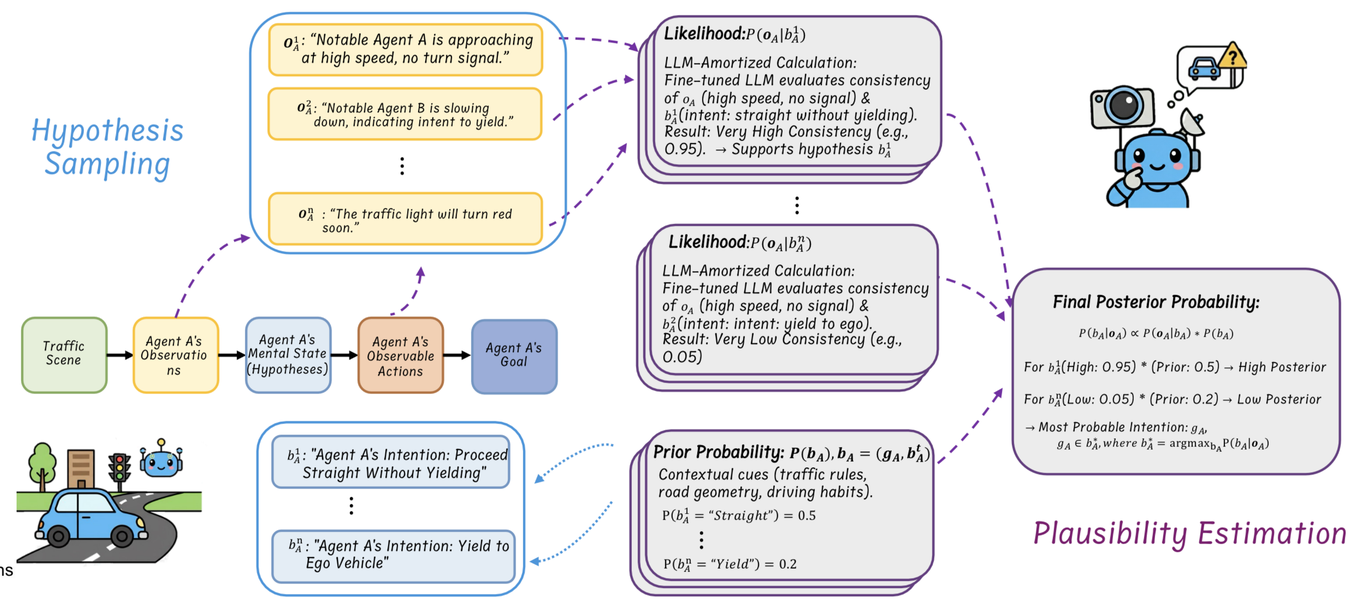
[3] Jiageng Mao, Junjie Ye, Yuxi Qian, Marco Pavone, and Yue Wang. A language agent for autonomous driving. In First Conference on Language Modeling, 2024.

Methods and Materials

- A comparison between the traditional **end-to-end planning** method, **human-like planning** method and **ToM-Driver** method:



- The workflow of our **LLM-amortized plausibility estimation**:



Results

- **Open-Loop Results**:

Method	L2 (m) ↓				Collision (%) ↓			
	1s	2s	3s	Avg.	1s	2s	3s	Avg.
NMP [58]	-	-	2.31	-	-	-	1.92	-
SA-NMP [58]	-	-	2.05	-	-	-	1.59	-
FF [19]	0.55	1.20	2.54	1.43	0.06	0.17	1.07	0.43
EO [25]	0.67	1.36	2.78	1.60	0.04	0.09	0.88	0.33
UniAD [21]	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31
VAD [23]	0.41	0.70	1.06	0.72	0.04	0.43	1.15	0.54
GPT-Driver [32]	0.27	0.74	1.52	0.84	0.07	0.15	1.10	0.44
Agent-Driver [33]	**0.22**	0.65	1.34	0.74	**0.02**	0.13	0.48	0.21
VLM-AD [55]	0.30	0.54	0.80	0.55	0.11	0.15	**0.38**	0.21
ReAL-AD [30]	0.35	0.53	0.71	0.53	0.09	0.12	0.31	**0.17**
ToM-Driver (Ours)	0.23	**0.50**	**0.81**	**0.51**	**0.02**	**0.11**	0.39	**0.17**

- **Stability** of ToM-Driver:

Percentage of training samples	0.10%	1%	10%	50%	100%
Number of invalid outputs	4	0	0	0	0

- **Closed-Loop Results**:

Method	DS ↑	RC ↑
CILRS [11]	7.47	13.40
LBC [12]	30.97	55.01
Transfuser [10]	54.52	78.41
ST-P3 [20]	55.14	86.74
VAD [23]	64.29	87.26
Agent-Driver [33]	57.33	91.37
ToM-Driver(Ours)	**66.17**	**92.03**

DS: Driving Score, RC: Route Completion

- **Ablation study** of ToM-Driver:

Method	L2 (m) ↓				Collision (%) ↓			
	1s	2s	3s	Avg.	1s	2s	3s	Avg.
Baseline	0.44	0.73	1.59	0.87	0.08	0.19	0.56	0.28
nuScenes-ToM Only	0.43	0.71	1.41	0.85	0.07	0.17	0.51	0.25
ToM Module Only	0.51	0.84	1.79	1.05	0.14	0.29	0.71	0.38
ToM-Driver	**0.23**	**0.50**	**0.81**	**0.51**	**0.02**	**0.11**	**0.39**	**0.17**

- **Plausibility Analysis** of the nuScenes-ToM dataset:

Sample Size (N)	100	500	1000	2000
Plausible number	100	499	996	1987
Plausible Rate (%)	100.0%	99.8%	99.6%	99.4%